UNIT 8 · STANDARD 6.AT.8

Solve and Graph Inequalities

Lesson 8-6

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve an inequality and graph its solution set on a number line.

Language Objective: I can explain my work using the words solve, graph, solution, and substitute.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Solve · **Graph** · **Solution** · **Substitute** · **Solution set** · **Inverse operation**

Level 3: try the blanks from memory first, then check the bank.

1. To find all the values that make an inequality true is to _____ it.

2. To show the solutions on a number line is to _____ them.

3. A value that makes an inequality true is a _____.

4. To check a solution, I _____ the value back into the inequality.

5 All the values that make an inequality true form the _____ .

6 An operation that undoes another to isolate the variable is an _____ .

Write About the Math

How much more does the detective need? How did you solve and graph this inequality?
¿Cuánto más necesita la detective? ¿Cómo resolviste y graficaste esta desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Solve** *Resolver*
- ☐ **Graph** *Graficar*
- ☐ **Solution** *Solución*
- ☐ **Substitute** *Sustituir*
- ☐ **Solution set** *Conjunto solución*

Model: *Solving an inequality gives a range of solutions, not one number.* — Students note both use inverse operations to isolate the variable, but an inequality's solution is many values (a range) while an equation's is a single value.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The detective needs at least $__ more because 48 + s ≥ 75 means s ≥ 75 − ___ = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.